

**CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS**

H1 BIOLOGY

8875/01

Paper 1 Multiple Choice

12 September 2008 Friday

1 hour

Additional materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or 2B recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the answer sheet in the spaces provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

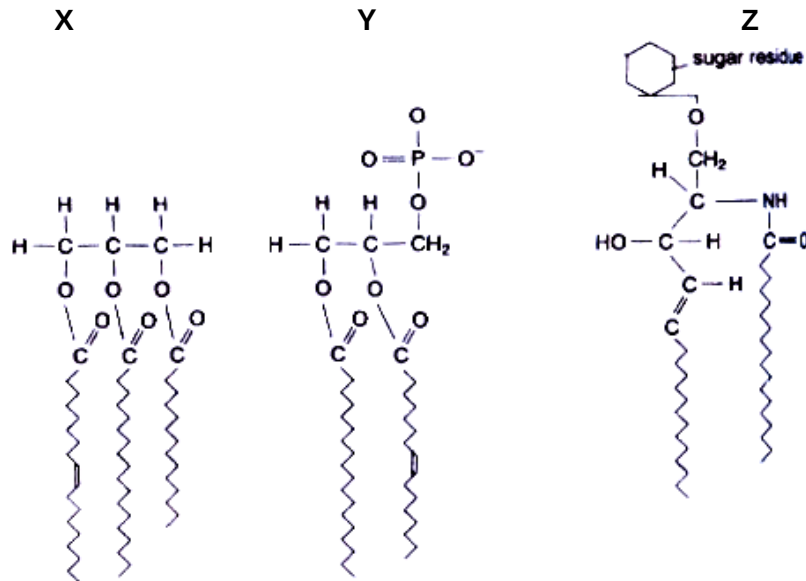
Read the instructions on the answer sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **10** printed pages.

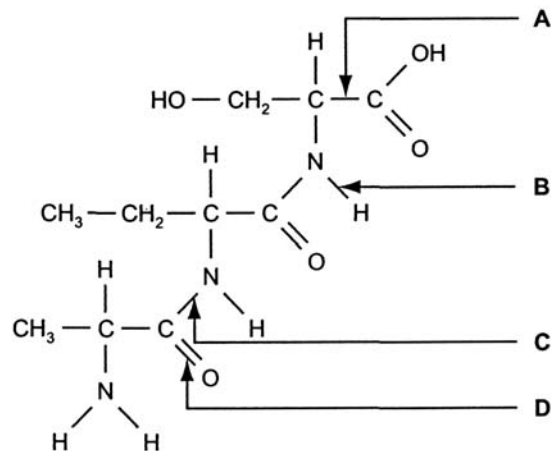
[Turn over

- 1 What is the main function of the Golgi apparatus?
- A formation of glycoprotein
 - B phosphorylation of ADP
 - C synthesis of proteins
 - D transport of RNA to the nucleus
- 2 The following diagram shows three molecules. Which best identifies them?

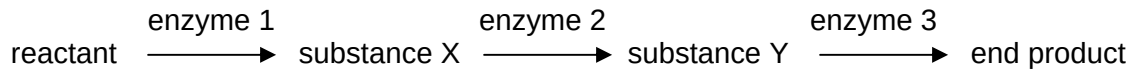


- | X | Y | Z |
|----------------|-----------------|--------------|
| A fatty acid | phospholipid | polypeptide |
| B fatty acid | triglyceride | glycoprotein |
| C triglyceride | phospholipid | glycolipid |
| D triglyceride | phosphoric acid | nucleic acid |

- 3 The diagram shows a molecule. Which arrow labels a peptide bond?

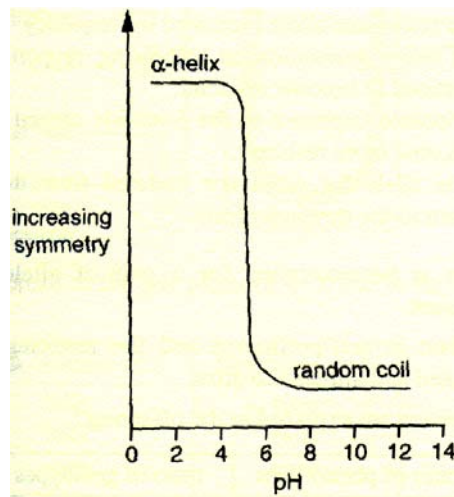


- 4 A metabolic pathway is shown below.



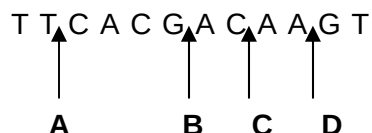
What would be the effect of adding a small amount of a non-competitive inhibitor of enzyme 2?

- A** Enzyme 2 would be partially denatured.
B Substance X would increase in concentration.
C Substance Y would no longer be formed.
D The initial reactant would no longer be metabolised.
- 5 Which statement describes events during interphase of mitosis?
- A** Chromosomes start to coil, condense and become visible.
B Nucleolus starts to disorganize.
C Semi-conservative replication takes place.
D Spindle fibres attach to kinetochore complex.
- 6 The graph shows the effect of pH on the structure of a protein which consist entirely of repeating units of one amino acid.



Which statement is true?

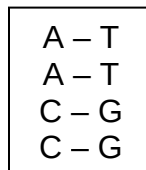
- A** At high acidity, the protein loses its secondary structure.
B At high acidity, the protein loses its tertiary structure.
C At low acidity, the protein loses its primary structure.
D At low acidity, the protein loses its secondary structure.
- 7 One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4 nm long.
What is the approximate length of the DNA coding sequence of a protein consisting of 111 amino acids?
- A** 132 nm
B 113 nm
C 66 nm
D 44 nm
- 8 The mRNA triplet UAG acts as a stop codon to terminate the synthesis of a polypeptide. The diagram shows a strand of DNA coding for 4 amino acids. Where would a mutation, involving the insertion of a thymine nucleotide, result in the termination of transcription?



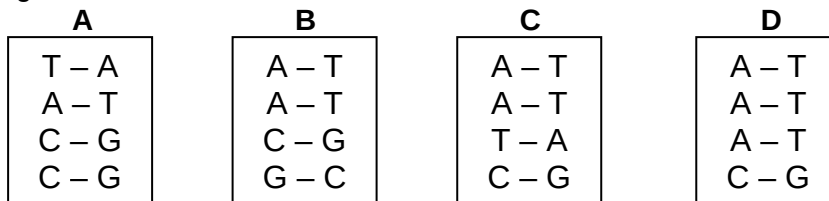
- 9 The table describes different kinds of mutations in DNA.

Mutation	Name
From purine to other purine	Transition
From pyrimidine to other pyrimidine	Transition
From purine to pyrimidine	Transversion
From pyrimidine to purine	Transversion

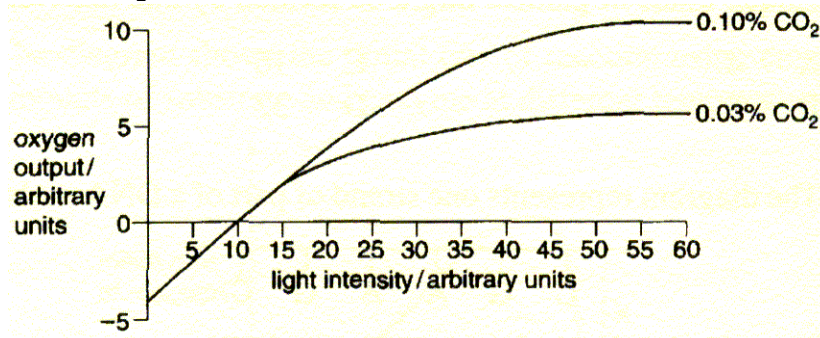
The diagram represents part of a DNA molecule.



Which diagram shows the DNA molecule after a transition has occurred?



- 10 The graph shows the relationship between oxygen production in photosynthesis and light intensity for the same green plant at two CO₂ concentrations.



Which of the following conclusions may be drawn from the graph?

- A Maximum photosynthetic rate at 0.03% CO₂ is lower than 0.10% because light intensity is limiting.
 B Oxygen output is negative when light intensity is lower than 10 units due to photorespiration.
 C Maximum photosynthetic rate at 0.10% CO₂ is higher because there is a higher chloroplast concentration.
 D The respiration rates at both CO₂ concentrations are the same.
- 11 If light intensity is uniformly high, under what conditions would carbon dioxide be most likely the limiting factor?

	Temperature/°C	% of carbon dioxide
A	25	4.00
B	20	0.04
C	10	0.04
D	5	0.01

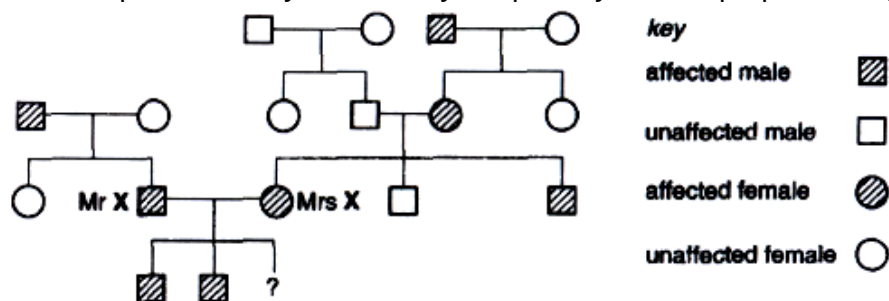
- 12 Intact mitochondria are mixed with a solution containing DCPIP, succinate and sucrose. The enzyme succinate dehydrogenase in the mitochondria oxidizes the succinate and the DCPIP is a hydrogen acceptor. What is the role of sucrose?
- A to act as a hydrogen donor
 - B to act as a reducing agent
 - C to maintain osmotic balance
 - D to provide energy for the reaction
- 13 Where are the enzymes involved in the chemical reactions which occur during glycolysis located?
- A in the fluid matrix of cytoplasm
 - B in the mitochondrial matrix
 - C in the nuclear sap
 - D on the cristae of a mitochondrion
- 14 In a small mammal, the allele for grey fur, **G**, is dominant to that for white fur, **g**. The allele for long tail, **T**, is dominant to the allele for short tail, **t**. Animals with grey fur and long tails were crossed with those having white fur and short tails.

The table shows the phenotypes of the 55 offspring.

number of offspring	fur	tail
15	grey	long
14	grey	short
14	white	long
12	white	short

What were the genotypes of the parents?

- A GgTt x gg tt
 - B GGTt x Gg tt
 - C GgTT x Gg tt
 - D GgTT x gg tt
- 15 The family tree was assembled by a genetic counsellor for Mr and Mrs X who suffer from Marfan's Syndrome, resulting in serious defects in heart valves and lungs. Children who inherit the dominant mutant allele from both parents rarely survive beyond puberty without proper management.



What is the probability that Mr and Mrs X's third child will be affected?

- A 0.75
- B 0.50
- C 0.25
- D 0.00

- 16 In guinea pigs, the allele **R** for rough coat is dominant over the allele **r** for smooth coat and the allele **B** for black fur is dominant over the allele **b** for white fur. The genes for fur colour and texture are not linked.

Two guinea pigs with genotype **RrBb** were mated together and one of the offspring had a rough, black coat. What is the probability that this offspring was homozygous for both rough coat and black fur?

- A 1 in 3
- B 1 in 9
- C 2 in 3
- D 2 in 9

- 17 Two genes, Q and R affect the size of petals of a flower.

Gene Q has two alleles, Q^L and Q^A . The genotype $Q^L Q^L$ produces large petals, $Q^L Q^A$ produces small petals and $Q^A Q^A$ produces no petals.

Gene R has two alleles. R produces red pigment, and dominant over r that produces no pigment. Two plants, heterozygous for both genes are crossed.

How many phenotypes are expected in the next generation?

- A 4
- B 6
- C 9
- D 12

- 18 The allele for an enzyme involved in the production of chlorophyll in geraniums is C^g . The mutant allele C^w codes for a defective gene with no pigment. A cross between two heterozygotes produced pale green plants and dark green plants in the ratio 2:1.

What is the most likely explanation for this ratio?

- A C^g is a dominant allele.
- B C^w is a dominant allele.
- C $C^w C^g$ is a lethal phenotype.
- D $C^w C^w$ is a lethal phenotype.

- 19 Four bacteriophage genes were mapped by recombination. The recombination frequencies between the four genes are shown in the table below.

	L	M	N	P
L	--	0.18	0.10	0.20
M		--	0.15	0.37
N			--	0.24
P				--

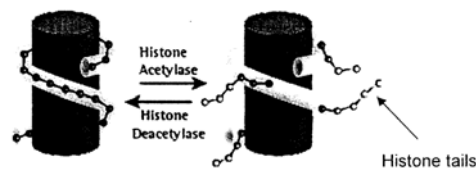
What is the order of the genes along the chromosome?

- A L, M, N, P
- B N, M, P, L
- C M, L, P, N
- D P, L, N, M

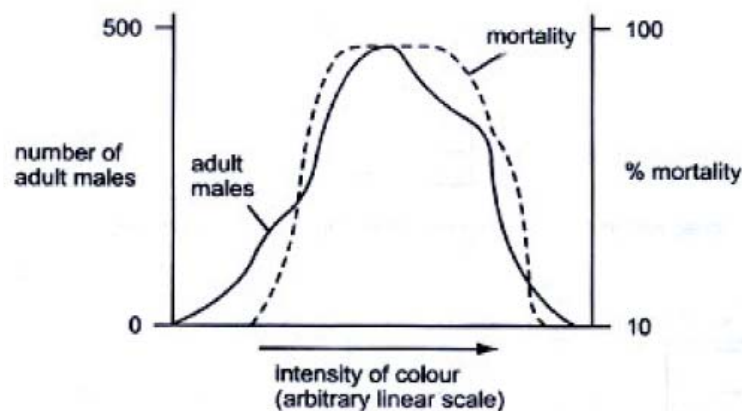
- 20 A mutation that renders the regulatory gene non-functional for the Lac operon would result in;

- A Continuous production of beta galactosidase, regardless of concentration of lactose.
- B Irreversible binding of repressor protein to the operator of Lac operon.
- C Complete blocking of the attachment site for RNA polymerase to the promoter.
- D No effect on transcription rate, as operon is inducible.

- 21 As shown in the diagram below, acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression. Suggest how this works.



- A Helicase action is enhanced by acetylation.
 B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
 C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
 D RNA polymerase works better by binding with acetyl groups.
- 22 Which of the following is a method of post-translational control in eukaryotic cells?
 A selectively degrading the proteins when they are transported to targeted destinations.
 B preventing the attachment of ribosomes.
 C inactivates essential translational factors.
 D selectively degrading the mRNA transcripts.
- 23 Which group can be considered a single population?
 A All the animals and plants on an isolated island.
 B All the birds counted in one day in a garden.
 C All the bacteria in a colony of *Bacillus subtilis*.
 D All insects occupying three hectares of farmland.
- 24 The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity. Which conclusion can be made from this graph?



- A Colour variation is environmentally induced.
 B Colour variation is genetically determined.
 C Extreme forms are favoured by natural selection.
 D The species shows discontinuous variation with respect to colour.

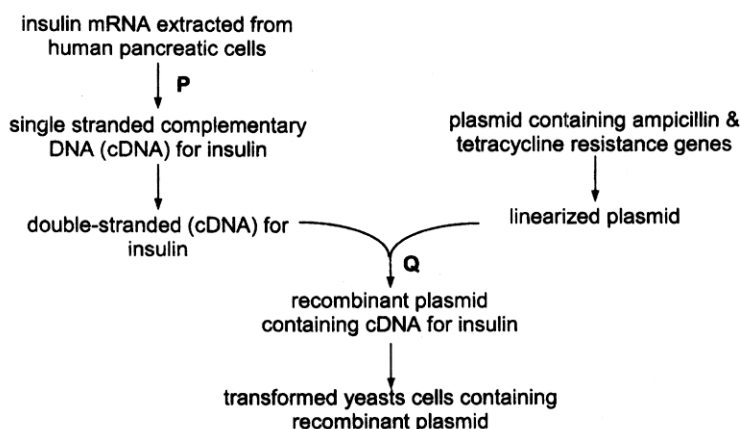
- 25 An insect collection contains 102 specimens of a species of butterfly. This specimen is sexually dimorphic, meaning that the males and females look different from each other. A student examined the specimens and collected the following data.

Observation	Frequency
Blue wing colour (male)	64
Brown wing colour (female)	38
Wing span 35-37 mm	15
Wing span 37-39 mm	68
Wing span 39-41 mm	19

How should this variation be classified?

	Continuous variation	Discontinuous variation
A	Sexual dimorphism	Colour
B	Colour	Sexual dimorphism
C	Sexual dimorphism	Wingspan
D	Wingspan	Sexual dimorphism

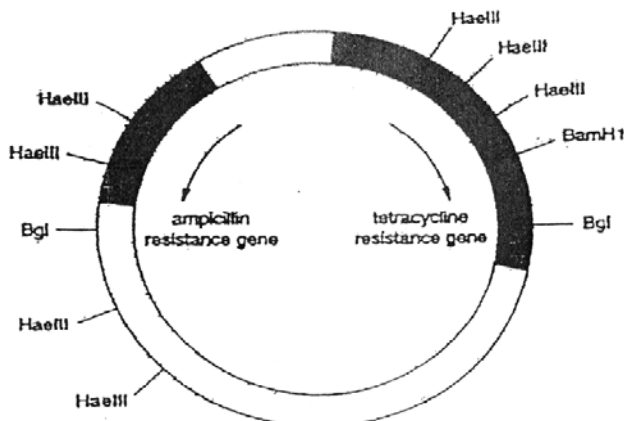
- 26 The following figure summarizes the formation of insulin using genetically engineered yeasts.



Which enzyme would be used at **P** and **Q**?

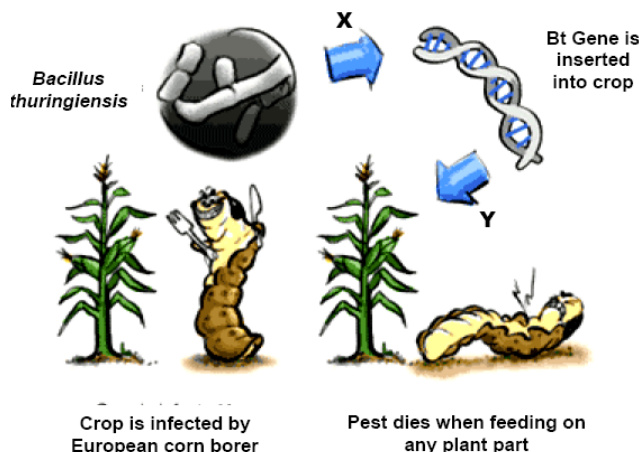
- | | | |
|---|-----------------------|--------------------|
| | P | Q |
| A | RNA polymerase | Restriction enzyme |
| B | Reverse transcriptase | Restriction enzyme |
| C | RNA polymerase | DNA ligase |
| D | Reverse transcriptase | DNA ligase |
- 27 Which of the following best describes a plasmid?
- A A gene within the chromosome.
- B Small circular piece of DNA outside the chromosome.
- C The genetic material of a bacteriophage.
- D A single, linear strand of DNA.

- 28 The gene map of a plasmid, which has been used for genetic engineering, is shown below. It contains two antibiotic resistance genes. The positions of several restriction endonuclease binding sites are also shown below.



Which enzyme(s) would be the most suitable for the cleavage of the plasmid and the DNA containing the gene of interest?

- A BamHI
B BglI
C HaellI
D BamHI and HaellI
- 29 The figure below illustrates how biotechnologists can make corn become resistant to infection by European corn borer.



Which of the following describe X and Y?

	X	Y
A	Bt Gene is isolated from <i>Bacillus thuringiensis</i> with the help of restriction endonucleases.	Bt Gene is inserted into crop via electrophoresis.
B	Bt Gene is isolated from <i>Bacillus thuringiensis</i> with the help of restriction endonucleases.	Bt Gene is inserted into crop via electroporation.
C	Bt Gene is isolated from <i>Bacillus thuringiensis</i> with the help of restriction exonucleases.	Bt Gene is inserted into crop as a result of infection by <i>Agrobacterium tumefaciens</i> .
D	Bt Gene is isolated from <i>Bacillus thuringiensis</i> with the help of restriction exonucleases.	Bt Gene is inserted into crop as a result of infection by <i>Bacillus thuringiensis</i> .

- 30** In one form of stem cell research the DNA from the cells of a patient is introduced into an empty ovum which is then induced to begin early embryonic development. Stem cells are harvested from this embryo to produce new healthy tissue for the original DNA donor. This is known as:
- A** therapeutic stem cell cloning
 - B** embryonic stem cell therapy
 - C** adult stem cell therapy
 - D** genetic engineering